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B.Tech. Degree IV Semester Examination April 2018

ME 15-1403 MECHATRONICS (2015 Scheme)

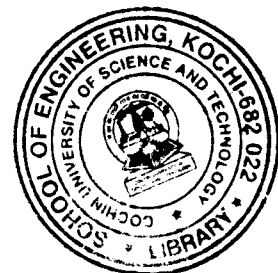
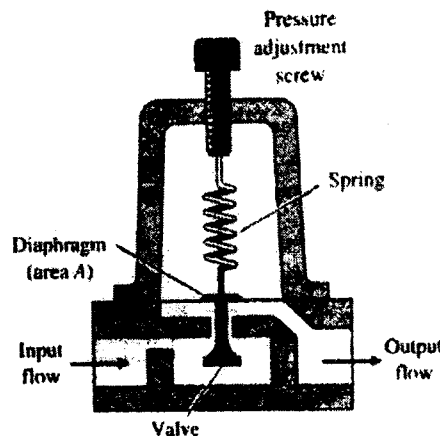
Time : 3 Hours

Maximum Marks : 60

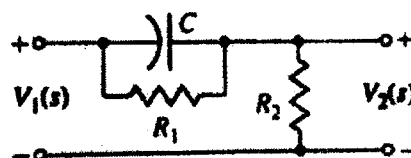
PART A (Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Describe the working of thermistors and RTDs. What are their relative merits and demerits?
- (b) Describe the working of an incremental encoder. Explain how the encoder detects direction of rotation.
- (c) A cutaway view of a commonly used pressure regulator is shown in the figure. The desired pressure is set by turning a calibrated screw. This compresses the spring and sets up a force that opposes the upward motion of the diaphragm. The bottom side of the diaphragm is exposed to the water pressure that is to be controlled. Thus the motion of the diaphragm is an indication of the pressure difference between the desired and the actual pressures. It acts like a comparator. The valve is connected to the diaphragm and moves according to the pressure difference until it reaches a position in which the difference is zero. Sketch a block diagram showing the control system with the output pressure as the regulated variable.

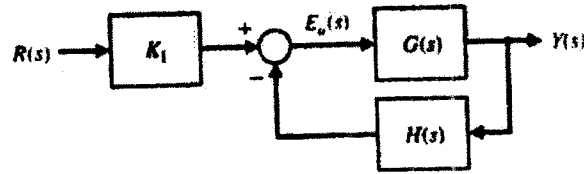


- (d) Describe a proportional plus integral controller.
- (e) A differentiating circuit is shown in the figure. Calculate the transfer function $V_2(s)/V_1(s)$.



(P.T.O.)

- (f) For the closed loop system given in the figure $G(s) = 40/(s+5)$, $H(s) = 20/(s+10)$ and $K_1 = 2$. Determine the steady-state error.

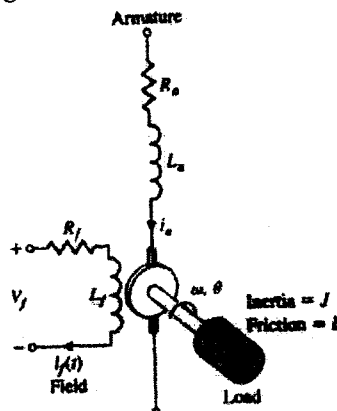


- (g) Determine the stability of the system represented by the characteristic equation $s^5 + 2s^4 + 2s^3 + 4s^2 + 11s + 10 = 0$ using Routh-Hurwitz criteria.
- (h) Explain full-stepping and half-stepping in a stepper motor.
- (i) Explain how the S-R flip-flop functions. Mention one use of flip-flop.
- (j) How are stack memory and stack pointers used while handling interrupts?

PART B

(4 × 10 = 40)

- II. (a) A thyristor is connected across a 120 V a.c supply. A gate current is given so that forward break down occurs at 60 V. After break down the voltage drops to 2 V. Draw the voltage across the thyristor for two cycles of the a.c. (3)
- (b) With neat sketch explain how a piezoelectric sensor functions. Mention two applications. (2)
- (c) A single acting pneumatic cylinder positioned vertically is used to lift weights. It is controlled by a three-port two-position direction control valve which has a solenoid on one side and a spring on the other. The cylinder should raise the weight when the solenoid is charged; and when the solenoid is off, the weight is lowered due to gravity. Draw the hydraulic circuit diagram for the case. (5)
- OR
- III. (a) Explain the working of a gear pump. (2)
- (b) Describe, with a neat diagram, how a 4-port, 3-position valve controls the flow to a double acting hydraulic cylinder. Draw the symbol for this valve. (4)
- (c) Explain pulse width modulation. Draw the circuit diagram and explain how the speed and direction of rotation of a d.c. motor is controlled using PWM signal. (4)
- IV. (a) Obtain the transfer function of the field controlled d.c. motor whose wiring diagram is shown in the figure. (5)

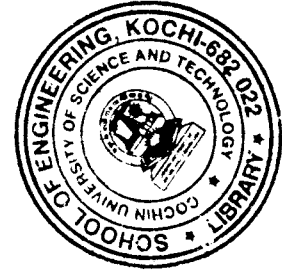
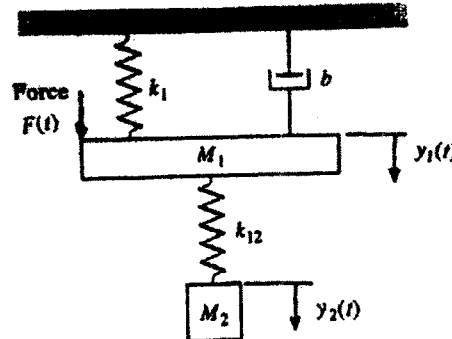


- (b) The transfer function of a system is $\frac{Y(s)}{R(s)} = \frac{10(s+2)}{s^2 + 8s + 15}$. Determine $y(t)$ when $r(t)$ is a unit step input. (5)

OR

(Contd..3.)

- V. (a) Obtain the differential equations describing the dynamic vibration absorber shown in the figure. This system is representative of many situations involving the vibration of machines containing unbalanced components. The parameters M_2 and k_{12} may be chosen so that the main mass M_1 does not vibrate when $F(t) = a \sin \omega_0 t$. Also sketch the analogous electrical circuit based on the force-current analogy. (5)



- (b) The closed-loop transfer function of the system is (5)
- $$T(s) = \frac{Y(s)}{R(s)} = \frac{2s^2 + 8s + 6}{s^3 + 8s^2 + 16s + 6}$$
- Obtain the state-space model.

- VI. Plot the Bode diagram for the following loop transfer function and determine the gain when the frequency is 10 rad/s. (10)

$$G(s)H(s) = \frac{2.5}{s(s^2 + 3s + 25)}$$

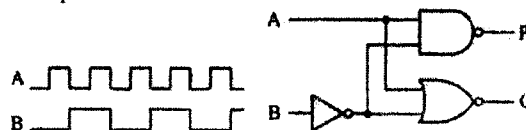
OR

- VII. Sketch root locus for the unity feedback control system having open loop transfer function $G(s) = \frac{K}{s(s+1)(s+2)}$. What is the range of values of K for which the system is stable? (10)

- VIII. (a) A washing machine has a pump to pump in water, a heater to heat water and another pump to pump water out. Write the ladder program to achieve the following: The pump is switched on to fill water for 3 minutes. Then the water is heated till the required temperature is reached, i.e., till the temperature sensor produce signal. On pressing the switch for the second pump, it pumps out water for 2 minutes and stops. (6)
- (b) Explain how a micro-processor executes an *odd* operation. (4)

OR

- IX. (a) The figure shows the input signals A and B and the gate system to which it is applied. Draw the output waveforms P and Q. (3)



- (b) Draw the ladder diagram and explain how the latch circuit works. (2)
- (c) Explain how an electronically controlled automobile engine is different from a conventional engine. What are its advantages? (5)
